

>> MainFunction

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[150*sin(q2)*sin(q1 - pi/2)*sin(q4 - pi/2) - 150*cos(q4 - pi/2)*(cos(q1 - pi/2)*sin
(q3 - pi/2) + cos(q2)*cos(q3 - pi/2)*sin(q1 - pi/2)) - 185*sin(q2)*sin(q1 - pi/2),
185*cos(q2)*cos(q1 - pi/2) - 150*cos(q2)*cos(q1 - pi/2)*sin(q4 - pi/2) - 150*cos(q1 -
pi/2)*cos(q3 - pi/2)*cos(q4 - pi/2)*sin(q2), -150*cos(q4 - pi/2)*(cos(q3 - pi/2)*sin
(q1 - pi/2) + cos(q2)*cos(q1 - pi/2)*sin(q3 - pi/2)), 150*sin(q4 - pi/2)*(sin(q1 -
pi/2)*sin(q3 - pi/2) - cos(q2)*cos(q1 - pi/2)*cos(q3 - pi/2)) - 150*cos(q1 - pi/2)
*cos(q4 - pi/2)*sin(q2)]
[185*cos(q1 - pi/2)*sin(q2) - 150*cos(q4 - pi/2)*(sin(q1 - pi/2)*sin(q3 - pi/2) - cos
(q2)*cos(q1 - pi/2)*cos(q3 - pi/2)) - 150*cos(q1 - pi/2)*sin(q2)*sin(q4 - pi/2),
185*cos(q2)*sin(q1 - pi/2) - 150*cos(q2)*sin(q1 - pi/2)*sin(q4 - pi/2) - 150*cos(q3 -
pi/2)*cos(q4 - pi/2)*sin(q2)*sin(q1 - pi/2), 150*cos(q4 - pi/2)*(cos(q1 - pi/2)*cos
(q3 - pi/2) - cos(q2)*sin(q1 - pi/2)*sin(q3 - pi/2)), -150*sin(q4 - pi/2)*(cos(q1 -
pi/2)*sin(q3 - pi/2) + cos(q2)*cos(q3 - pi/2)*sin(q1 - pi/2)) - 150*cos(q4 - pi/2)
*sin(q2)*sin(q1 - pi/2)]
[
0,
150*sin(q2)*sin(q4 - pi/2) - 185*sin
(q2) - 150*cos(q2)*cos(q3 - pi/2)*cos(q4 - pi/2),
150*cos(q4 - pi/2)*sin(q2)*sin(q3 - pi/2),
150*cos(q3 - pi/2)*sin(q2)*sin(q4 - pi/2) - 150*cos(q2)*cos(q4 - pi/2)]

[150*sin(q2)*sin(q1 - pi/2)*sin(q4 - pi/2) - 150*cos(q4 - pi/2)*(cos(q1 - pi/2)*sin
(q3 - pi/2) + cos(q2)*cos(q3 - pi/2)*sin(q1 - pi/2)) - 185*sin(q2)*sin(q1 - pi/2),
-cos(q1 - pi/2)*(150*cos(q2)*sin(q4 - pi/2) - 185*cos(q2) + 150*cos(q3 - pi/2)*cos(q4
- pi/2)*sin(q2)),
- cos(q2)*(185*sin(q2)*sin(q1 - pi/2) + 150*cos(q4 - pi/2)*(cos(q1 - pi/2)*sin(q3 -
pi/2) + cos(q2)*cos(q3 - pi/2)*sin(q1 - pi/2)) - 150*sin(q2)*sin(q1 - pi/2)*sin(q4 -
pi/2)) - sin(q2)*sin(q1 - pi/2)*(150*cos(q2)*sin(q4 - pi/2) - 185*cos(q2) + 150*cos
(q3 - pi/2)*cos(q4 - pi/2)*sin(q2)),
- (150*cos(q2)*sin(q4 - pi/2) + 150*cos(q3 - pi/2)*cos(q4 - pi/2)*sin(q2))*(cos(q1 -
pi/2)*cos(q3 - pi/2) - cos(q2)*sin(q1 - pi/2)*sin(q3 - pi/2)) - sin(q2)*sin(q3 -
pi/2)*(150*cos(q4 - pi/2)*(cos(q1 - pi/2)*sin(q3 - pi/2) + cos(q2)*cos(q3 - pi/2)*sin
(q1 - pi/2)) - 150*sin(q2)*sin(q1 - pi/2)*sin(q4 - pi/2))]
[185*cos(q1 - pi/2)*sin(q2) - 150*cos(q4 - pi/2)*(sin(q1 - pi/2)*sin(q3 - pi/2) - cos
(q2)*cos(q1 - pi/2)*cos(q3 - pi/2)) - 150*cos(q1 - pi/2)*sin(q2)*sin(q4 - pi/2),
-sin(q1 - pi/2)*(150*cos(q2)*sin(q4 - pi/2) - 185*cos(q2) + 150*cos(q3 - pi/2)*cos(q4
- pi/2)*sin(q2)),
cos(q1 - pi/2)*sin(q2)*(150*cos(q2)*sin(q4 - pi/2) - 185*cos(q2) + 150*cos(q3 - pi/2)
*cos(q4 - pi/2)*sin(q2)) - cos(q2)*(150*cos(q4 - pi/2)*(sin(q1 - pi/2)*sin(q3 - pi/2)
- cos(q2)*cos(q1 - pi/2)*cos(q3 - pi/2)) - 185*cos(q1 - pi/2)*sin(q2) + 150*cos(q1 -
pi/2)*sin(q2)*sin(q4 - pi/2)),
- (150*cos(q2)*sin(q4 - pi/2) + 150*cos(q3 - pi/2)*cos(q4 - pi/2)*sin(q2))*(cos(q3 -
pi/2)*sin(q1 - pi/2) + cos(q2)*cos(q1 - pi/2)*sin(q3 - pi/2)) - sin(q2)*sin(q3 -
pi/2)*(150*cos(q4 - pi/2)*(sin(q1 - pi/2)*sin(q3 - pi/2) - cos(q2)*cos(q1 - pi/2)*cos
(q3 - pi/2)) + 150*cos(q1 - pi/2)*sin(q2)*sin(q4 - pi/2))]
[
0, cos(q1 - pi/2)*(150*cos(q4 - pi/2)*(sin(q1 - pi/2)*sin(q3 - pi/2) - cos(q2)*cos(q1
- pi/2)*cos(q3 - pi/2)) - 185*cos(q1 - pi/2)*sin(q2) + 150*cos(q1 - pi/2)*sin(q2)*sin
(q4 - pi/2)) - sin(q1 - pi/2)*(185*sin(q2)*sin(q1 - pi/2) + 150*cos(q4 - pi/2)*(cos
(q1 - pi/2)*sin(q3 - pi/2) + cos(q2)*cos(q3 - pi/2)*sin(q1 - pi/2)) - 150*sin(q2)*sin
(q1 - pi/2)*sin(q4 - pi/2)), sin(q2)*sin(q1 - pi/2)*(150*cos(q4 - pi/2)*(sin(q1 -
pi/2)*sin(q3 - pi/2) - cos(q2)*cos(q1 - pi/2)*cos(q3 - pi/2)) - 185*cos(q1 - pi/2)
*sin(q2) + 150*cos(q1 - pi/2)*sin(q2)*sin(q4 - pi/2)) + cos(q1 - pi/2)*sin(q2)*
(185*sin(q2)*sin(q1 - pi/2) + 150*cos(q4 - pi/2)*(cos(q1 - pi/2)*sin(q3 - pi/2) + cos
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(q2)*cos(q3 - pi/2)*sin(q1 - pi/2)) - 150*sin(q2)*sin(q1 - pi/2)*sin(q4 - pi/2)),
(150*cos(q4 - pi/2)*(sin(q1 - pi/2)*sin(q3 - pi/2) - cos(q2)*cos(q1 - pi/2)*cos(q3 -
pi/2)) + 150*cos(q1 - pi/2)*sin(q2)*sin(q4 - pi/2))*(cos(q1 - pi/2)*cos(q3 - pi/2) -
cos(q2)*sin(q1 - pi/2)*sin(q3 - pi/2)) - (150*cos(q4 - pi/2)*(cos(q1 - pi/2)*sin(q3 -
pi/2) + cos(q2)*cos(q3 - pi/2)*sin(q1 - pi/2)) - 150*sin(q2)*sin(q1 - pi/2)*sin(q4 -
pi/2))*(cos(q3 - pi/2)*sin(q1 - pi/2) + cos(q2)*cos(q1 - pi/2)*sin(q3 - pi/2))]
[
0,
-sin(q1 - pi/2),
cos(q1 - pi/2)*sin(q2),
- cos(q3 - pi/2)*sin(q1 - pi/2) - cos(q2)*cos(q1 - pi/2)*sin(q3 - pi/2)]
[
0,
cos(q1 - pi/2),
sin(q2)*sin(q1 - pi/2),
cos(q1 - pi/2)*cos(q3 - pi/2) - cos(q2)*sin(q1 - pi/2)*sin(q3 - pi/2)]
[
1,
0,
cos(q2),
sin(q2)*sin(q3 - pi/2)]

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[q4dot*(150*sin(q4 - pi/2)*(cos(q1 - pi/2)*sin(q3 - pi/2) + cos(q2)*cos(q3 - pi/2)
*sin(q1 - pi/2)) + 150*cos(q4 - pi/2)*sin(q2)*sin(q1 - pi/2)) + q1dot*(150*cos(q4 -
pi/2)*(sin(q1 - pi/2)*sin(q3 - pi/2) - cos(q2)*cos(q1 - pi/2)*cos(q3 - pi/2)) -
185*cos(q1 - pi/2)*sin(q2) + 150*cos(q1 - pi/2)*sin(q2)*sin(q4 - pi/2)) + q2dot*
(150*cos(q2)*sin(q1 - pi/2)*sin(q4 - pi/2) - 185*cos(q2)*sin(q1 - pi/2) + 150*cos(q3 -
pi/2)*cos(q4 - pi/2)*sin(q2)*sin(q1 - pi/2)) - 150*q3dot*cos(q4 - pi/2)*(cos(q1 -
pi/2)*cos(q3 - pi/2) - cos(q2)*sin(q1 - pi/2)*sin(q3 - pi/2)),
q1dot*sin(q1 - pi/2)*(150*cos(q2)*sin(q4 - pi/2) - 185*cos(q2) + 150*cos(q3 - pi/2)
*cos(q4 - pi/2)*sin(q2)) - q4dot*cos(q1 - pi/2)*(150*cos(q2)*cos(q4 - pi/2) - 150*cos
(q3 - pi/2)*sin(q2)*sin(q4 - pi/2)) - q2dot*cos(q1 - pi/2)*(185*sin(q2) - 150*sin(q2)
*sin(q4 - pi/2) + 150*cos(q2)*cos(q3 - pi/2)*cos(q4 - pi/2)) + 150*q3dot*cos(q1 -
pi/2)*cos(q4 - pi/2)*sin(q2)*sin(q3 - pi/2),
q4dot*(cos(q2)*(150*sin(q4 - pi/2)*(cos(q1 - pi/2)*sin(q3 - pi/2) + cos(q2)*cos(q3 - pi/2)*sin(q1 - pi/2)) +
150*cos(q4 - pi/2)*sin(q2)*sin(q1 - pi/2)) - sin(q2)*sin(q1 - pi/2)*(150*cos(q2)*cos
(q4 - pi/2) - 150*cos(q3 - pi/2)*sin(q2)*sin(q4 - pi/2))) - q3dot*(150*cos(q2)*cos(q4 -
pi/2)*(cos(q1 - pi/2)*cos(q3 - pi/2) - cos(q2)*sin(q1 - pi/2)*sin(q3 - pi/2)) -
150*cos(q4 - pi/2)*sin(q2)^2*sin(q1 - pi/2)*sin(q3 - pi/2)) + q1dot*(cos(q2)*(150*cos
(q4 - pi/2)*(sin(q1 - pi/2)*sin(q3 - pi/2) - cos(q2)*cos(q1 - pi/2)*cos(q3 - pi/2)) -
185*cos(q1 - pi/2)*sin(q2) + 150*cos(q1 - pi/2)*sin(q2)*sin(q4 - pi/2)) - cos(q1 -
pi/2)*sin(q2)*(150*cos(q2)*sin(q4 - pi/2) - 185*cos(q2) + 150*cos(q3 - pi/2)*cos(q4 -
pi/2)*sin(q2))) + q2dot*(cos(q2)*(150*cos(q2)*sin(q1 - pi/2)*sin(q4 - pi/2) - 185*cos
(q2)*sin(q1 - pi/2) + 150*cos(q3 - pi/2)*cos(q4 - pi/2)*sin(q2)*sin(q1 - pi/2)) + sin
(q2)*(185*sin(q2)*sin(q1 - pi/2) + 150*cos(q4 - pi/2)*(cos(q1 - pi/2)*sin(q3 - pi/2)
+ cos(q2)*cos(q3 - pi/2)*sin(q1 - pi/2)) - 150*sin(q2)*sin(q1 - pi/2)*sin(q4 - pi/2))
- sin(q2)*sin(q1 - pi/2)*(185*sin(q2) - 150*sin(q2)*sin(q4 - pi/2) + 150*cos(q2)*cos
(q3 - pi/2)*cos(q4 - pi/2)) - cos(q2)*sin(q1 - pi/2)*(150*cos(q2)*sin(q4 - pi/2) -
185*cos(q2) + 150*cos(q3 - pi/2)*cos(q4 - pi/2)*sin(q2))),
q1dot*((150*cos(q2)*sin(q4 - pi/2) + 150*cos(q3 - pi/2)*cos(q4 - pi/2)*sin(q2))*(cos
(q3 - pi/2)*sin(q1 - pi/2) + cos(q2)*cos(q1 - pi/2)*sin(q3 - pi/2)) + sin(q2)*sin(q3 -
pi/2)*(150*cos(q4 - pi/2)*(sin(q1 - pi/2)*sin(q3 - pi/2) - cos(q2)*cos(q1 - pi/2)
*cos(q3 - pi/2)) + 150*cos(q1 - pi/2)*sin(q2)*sin(q4 - pi/2))) + q3dot*((150*cos(q2)
*sin(q4 - pi/2) + 150*cos(q3 - pi/2)*cos(q4 - pi/2)*sin(q2))*(cos(q1 - pi/2)*sin(q3 -

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$$\begin{aligned}
& \pi/2) + \cos(q_2) \cos(q_3 - \pi/2) \sin(q_1 - \pi/2)) - \cos(q_3 - \pi/2) \sin(q_2) (150 \cos(q_4 - \pi/2) \\
& \pi/2) (\cos(q_1 - \pi/2) \sin(q_3 - \pi/2) + \cos(q_2) \cos(q_3 - \pi/2) \sin(q_1 - \pi/2)) - \\
& 150 \sin(q_2) \sin(q_1 - \pi/2) \sin(q_4 - \pi/2)) - q_4 \text{dot} ((150 \cos(q_2) \cos(q_4 - \pi/2) - \\
& 150 \cos(q_3 - \pi/2) \sin(q_2) \sin(q_4 - \pi/2)) (\cos(q_1 - \pi/2) \cos(q_3 - \pi/2) - \cos(q_2) \\
& \sin(q_1 - \pi/2) \sin(q_3 - \pi/2)) - \sin(q_2) \sin(q_3 - \pi/2) (150 \sin(q_4 - \pi/2) (\cos(q_1 \\
& - \pi/2) \sin(q_3 - \pi/2) + \cos(q_2) \cos(q_3 - \pi/2) \sin(q_1 - \pi/2)) + 150 \cos(q_4 - \pi/2) \\
& \sin(q_2) \sin(q_1 - \pi/2))) + q_2 \text{dot} ((150 \sin(q_2) \sin(q_4 - \pi/2) - 150 \cos(q_2) \cos(q_3 - \\
& \pi/2) \cos(q_4 - \pi/2)) (\cos(q_1 - \pi/2) \cos(q_3 - \pi/2) - \cos(q_2) \sin(q_1 - \pi/2) \sin(q_3 \\
& - \pi/2)) - \cos(q_2) \sin(q_3 - \pi/2) (150 \cos(q_4 - \pi/2) (\cos(q_1 - \pi/2) \sin(q_3 - \pi/2) \\
& + \cos(q_2) \cos(q_3 - \pi/2) \sin(q_1 - \pi/2)) - 150 \sin(q_2) \sin(q_1 - \pi/2) \sin(q_4 - \pi/2)) \\
& + \sin(q_2) \sin(q_3 - \pi/2) (150 \cos(q_2) \sin(q_1 - \pi/2) \sin(q_4 - \pi/2) + 150 \cos(q_3 - \\
& \pi/2) \cos(q_4 - \pi/2) \sin(q_2) \sin(q_1 - \pi/2)) - \sin(q_2) \sin(q_1 - \pi/2) \sin(q_3 - \pi/2) \\
& (150 \cos(q_2) \sin(q_4 - \pi/2) + 150 \cos(q_3 - \pi/2) \cos(q_4 - \pi/2) \sin(q_2))) \\
& [q_4 \text{dot} (150 \sin(q_4 - \pi/2) (\sin(q_1 - \pi/2) \sin(q_3 - \pi/2) - \cos(q_2) \cos(q_1 - \pi/2) \\
& \cos(q_3 - \pi/2)) - 150 \cos(q_1 - \pi/2) \cos(q_4 - \pi/2) \sin(q_2)) - q_2 \text{dot} (150 \cos(q_2) \\
& \cos(q_1 - \pi/2) \sin(q_4 - \pi/2) - 185 \cos(q_2) \cos(q_1 - \pi/2) + 150 \cos(q_1 - \pi/2) \cos \\
& (q_3 - \pi/2) \cos(q_4 - \pi/2) \sin(q_2)) - q_1 \text{dot} (185 \sin(q_2) \sin(q_1 - \pi/2) + 150 \cos(q_4 \\
& - \pi/2) (\cos(q_1 - \pi/2) \sin(q_3 - \pi/2) + \cos(q_2) \cos(q_3 - \pi/2) \sin(q_1 - \pi/2)) - \\
& 150 \sin(q_2) \sin(q_1 - \pi/2) \sin(q_4 - \pi/2)) - 150 q_3 \text{dot} \cos(q_4 - \pi/2) (\cos(q_3 - \pi/2) \\
& \sin(q_1 - \pi/2) + \cos(q_2) \cos(q_1 - \pi/2) \sin(q_3 - \pi/2)), \\
& 150 q_3 \text{dot} \cos(q_4 - \pi/2) \sin(q_2) \sin(q_1 - \pi/2) \sin(q_3 - \pi/2) - q_1 \text{dot} \cos(q_1 - \pi/2) \\
& (150 \cos(q_2) \sin(q_4 - \pi/2) - 185 \cos(q_2) + 150 \cos(q_3 - \pi/2) \cos(q_4 - \pi/2) \sin \\
& (q_2)) - q_2 \text{dot} \sin(q_1 - \pi/2) (185 \sin(q_2) - 150 \sin(q_2) \sin(q_4 - \pi/2) + 150 \cos(q_2) \\
& \cos(q_3 - \pi/2) \cos(q_4 - \pi/2)) - q_4 \text{dot} \sin(q_1 - \pi/2) (150 \cos(q_2) \cos(q_4 - \pi/2) - \\
& 150 \cos(q_3 - \pi/2) \sin(q_2) \sin(q_4 - \pi/2)), \quad q_2 \text{dot} (\sin(q_2) (150 \cos(q_4 - \pi/2) \\
& (\sin(q_1 - \pi/2) \sin(q_3 - \pi/2) - \cos(q_2) \cos(q_1 - \pi/2) \cos(q_3 - \pi/2)) - 185 \cos(q_1 \\
& - \pi/2) \sin(q_2) + 150 \cos(q_1 - \pi/2) \sin(q_2) \sin(q_4 - \pi/2)) - \cos(q_2) (150 \cos(q_2) \\
& \cos(q_1 - \pi/2) \sin(q_4 - \pi/2) - 185 \cos(q_2) \cos(q_1 - \pi/2) + 150 \cos(q_1 - \pi/2) \cos \\
& (q_3 - \pi/2) \cos(q_4 - \pi/2) \sin(q_2)) + \cos(q_2) \cos(q_1 - \pi/2) (150 \cos(q_2) \sin(q_4 - \\
& \pi/2) - 185 \cos(q_2) + 150 \cos(q_3 - \pi/2) \cos(q_4 - \pi/2) \sin(q_2)) + \cos(q_1 - \pi/2) \sin \\
& (q_2) (185 \sin(q_2) - 150 \sin(q_2) \sin(q_4 - \pi/2) + 150 \cos(q_2) \cos(q_3 - \pi/2) \cos(q_4 - \\
& \pi/2))) - q_3 \text{dot} (150 \cos(q_2) \cos(q_4 - \pi/2) (\cos(q_3 - \pi/2) \sin(q_1 - \pi/2) + \cos(q_2) \\
& \cos(q_1 - \pi/2) \sin(q_3 - \pi/2)) + 150 \cos(q_1 - \pi/2) \cos(q_4 - \pi/2) \sin(q_2)^2 \sin(q_3 \\
& - \pi/2)) - q_1 \text{dot} (\cos(q_2) (185 \sin(q_2) \sin(q_1 - \pi/2) + 150 \cos(q_4 - \pi/2) (\cos(q_1 - \\
& \pi/2) \sin(q_3 - \pi/2) + \cos(q_2) \cos(q_3 - \pi/2) \sin(q_1 - \pi/2)) - 150 \sin(q_2) \sin(q_1 - \\
& \pi/2) \sin(q_4 - \pi/2)) + \sin(q_2) \sin(q_1 - \pi/2) (150 \cos(q_2) \sin(q_4 - \pi/2) - 185 \cos \\
& (q_2) + 150 \cos(q_3 - \pi/2) \cos(q_4 - \pi/2) \sin(q_2))) + q_4 \text{dot} (\cos(q_2) (150 \sin(q_4 - \\
& \pi/2) (\sin(q_1 - \pi/2) \sin(q_3 - \pi/2) - \cos(q_2) \cos(q_1 - \pi/2) \cos(q_3 - \pi/2)) - \\
& 150 \cos(q_1 - \pi/2) \cos(q_4 - \pi/2) \sin(q_2)) + \cos(q_1 - \pi/2) \sin(q_2) (150 \cos(q_2) \cos \\
& (q_4 - \pi/2) - 150 \cos(q_3 - \pi/2) \sin(q_2) \sin(q_4 - \pi/2))), \\
& q_3 \text{dot} ((150 \cos(q_2) \sin(q_4 - \pi/2) + 150 \cos(q_3 - \pi/2) \cos(q_4 - \pi/2) \sin(q_2)) (\sin \\
& (q_1 - \pi/2) \sin(q_3 - \pi/2) - \cos(q_2) \cos(q_1 - \pi/2) \cos(q_3 - \pi/2)) - \cos(q_3 - \pi/2) \\
& \sin(q_2) (150 \cos(q_4 - \pi/2) (\sin(q_1 - \pi/2) \sin(q_3 - \pi/2) - \cos(q_2) \cos(q_1 - \pi/2) \\
& \cos(q_3 - \pi/2)) + 150 \cos(q_1 - \pi/2) \sin(q_2) \sin(q_4 - \pi/2))) - q_4 \text{dot} ((150 \cos(q_2) \\
& \cos(q_4 - \pi/2) - 150 \cos(q_3 - \pi/2) \sin(q_2) \sin(q_4 - \pi/2)) (\cos(q_3 - \pi/2) \sin(q_1 - \\
& \pi/2) + \cos(q_2) \cos(q_1 - \pi/2) \sin(q_3 - \pi/2)) - \sin(q_2) \sin(q_3 - \pi/2) (150 \sin(q_4 - \\
& \pi/2) (\sin(q_1 - \pi/2) \sin(q_3 - \pi/2) - \cos(q_2) \cos(q_1 - \pi/2) \cos(q_3 - \pi/2)) - \\
& 150 \cos(q_1 - \pi/2) \cos(q_4 - \pi/2) \sin(q_2))) - q_1 \text{dot} ((150 \cos(q_2) \sin(q_4 - \pi/2) + \\
& 150 \cos(q_3 - \pi/2) \cos(q_4 - \pi/2) \sin(q_2)) (\cos(q_1 - \pi/2) \cos(q_3 - \pi/2) - \cos(q_2) \\
& \sin(q_1 - \pi/2) \sin(q_3 - \pi/2)) + \sin(q_2) \sin(q_3 - \pi/2) (150 \cos(q_4 - \pi/2) (\cos(q_1 \\
& - \pi/2) \sin(q_3 - \pi/2) + \cos(q_2) \cos(q_3 - \pi/2) \sin(q_1 - \pi/2)) - 150 \sin(q_2) \sin(q_1 \\
& - \pi/2) \sin(q_4 - \pi/2))) + q_2 \text{dot} ((150 \sin(q_2) \sin(q_4 - \pi/2) - 150 \cos(q_2) \cos(q_3 - \\
& \pi/2) \cos(q_4 - \pi/2)) (\cos(q_3 - \pi/2) \sin(q_1 - \pi/2) + \cos(q_2) \cos(q_1 - \pi/2) \sin(q_3 \\
& - \pi/2)) - \cos(q_2) \sin(q_3 - \pi/2) (150 \cos(q_4 - \pi/2) (\sin(q_1 - \pi/2) \sin(q_3 - \pi/2) \\
\end{aligned}$$

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- cos(q2)*cos(q1 - pi/2)*cos(q3 - pi/2)) + 150*cos(q1 - pi/2)*sin(q2)*sin(q4 - pi/2))
- sin(q2)*sin(q3 - pi/2)*(150*cos(q2)*cos(q1 - pi/2)*sin(q4 - pi/2) + 150*cos(q1 -
pi/2)*cos(q3 - pi/2)*cos(q4 - pi/2)*sin(q2)) + cos(q1 - pi/2)*sin(q2)*sin(q3 - pi/2)*
(150*cos(q2)*sin(q4 - pi/2) + 150*cos(q3 - pi/2)*cos(q4 - pi/2)*sin(q2)))]
[
0, q3dot*(150*cos(q1 - pi/2)*cos(q4 - pi/2)*(cos(q3 - pi/2)*sin(q1 - pi/2) + cos(q2)
*cos(q1 - pi/2)*sin(q3 - pi/2)) - 150*cos(q4 - pi/2)*sin(q1 - pi/2)*(cos(q1 - pi/2)
*cos(q3 - pi/2) - cos(q2)*sin(q1 - pi/2)*sin(q3 - pi/2))) - q4dot*(cos(q1 - pi/2)*
(150*sin(q4 - pi/2)*(sin(q1 - pi/2)*sin(q3 - pi/2) - cos(q2)*cos(q1 - pi/2)*cos(q3 -
pi/2)) - 150*cos(q1 - pi/2)*cos(q4 - pi/2)*sin(q2)) - sin(q1 - pi/2)*(150*sin(q4 -
pi/2)*(cos(q1 - pi/2)*sin(q3 - pi/2) + cos(q2)*cos(q3 - pi/2)*sin(q1 - pi/2)) +
150*cos(q4 - pi/2)*sin(q2)*sin(q1 - pi/2))) + q2dot*(cos(q1 - pi/2)*(150*cos(q2)*cos
(q1 - pi/2)*sin(q4 - pi/2) - 185*cos(q2)*cos(q1 - pi/2) + 150*cos(q1 - pi/2)*cos(q3 -
pi/2)*cos(q4 - pi/2)*sin(q2)) + sin(q1 - pi/2)*(150*cos(q2)*sin(q1 - pi/2)*sin(q4 -
pi/2) - 185*cos(q2)*sin(q1 - pi/2) + 150*cos(q3 - pi/2)*cos(q4 - pi/2)*sin(q2)*sin(q1
- pi/2))), q2dot*(sin(q2)*sin(q1 - pi/2)*(150*cos(q2)*cos(q1 - pi/2)*sin(q4 - pi/2) -
185*cos(q2)*cos(q1 - pi/2) + 150*cos(q1 - pi/2)*cos(q3 - pi/2)*cos(q4 - pi/2)*sin
(q2)) + cos(q2)*sin(q1 - pi/2)*(150*cos(q4 - pi/2)*(sin(q1 - pi/2)*sin(q3 - pi/2) -
cos(q2)*cos(q1 - pi/2)*cos(q3 - pi/2)) - 185*cos(q1 - pi/2)*sin(q2) + 150*cos(q1 -
pi/2)*sin(q2)*sin(q4 - pi/2)) + cos(q2)*cos(q1 - pi/2)*(185*sin(q2)*sin(q1 - pi/2) +
150*cos(q4 - pi/2)*(cos(q1 - pi/2)*sin(q3 - pi/2) + cos(q2)*cos(q3 - pi/2)*sin(q1 -
pi/2)) - 150*sin(q2)*sin(q1 - pi/2)*sin(q4 - pi/2)) - cos(q1 - pi/2)*sin(q2)*(150*cos
(q2)*sin(q1 - pi/2)*sin(q4 - pi/2) - 185*cos(q2)*sin(q1 - pi/2) + 150*cos(q3 - pi/2)
*cos(q4 - pi/2)*sin(q2)*sin(q1 - pi/2))) - q4dot*(cos(q1 - pi/2)*sin(q2)*(150*sin(q4
- pi/2)*(cos(q1 - pi/2)*sin(q3 - pi/2) + cos(q2)*cos(q3 - pi/2)*sin(q1 - pi/2)) +
150*cos(q4 - pi/2)*sin(q2)*sin(q1 - pi/2)) + sin(q2)*sin(q1 - pi/2)*(150*sin(q4 -
pi/2)*(sin(q1 - pi/2)*sin(q3 - pi/2) - cos(q2)*cos(q1 - pi/2)*cos(q3 - pi/2)) -
150*cos(q1 - pi/2)*cos(q4 - pi/2)*sin(q2))) + q3dot*(150*cos(q1 - pi/2)*cos(q4 -
pi/2)*sin(q2)*(cos(q1 - pi/2)*cos(q3 - pi/2) - cos(q2)*sin(q1 - pi/2)*sin(q3 - pi/2))
+ 150*cos(q4 - pi/2)*sin(q2)*sin(q1 - pi/2)*(cos(q3 - pi/2)*sin(q1 - pi/2) + cos(q2)
*cos(q1 - pi/2)*sin(q3 - pi/2))), q2dot*((150*cos(q2)*cos(q1 - pi/2)*sin(q4 - pi/2) +
150*cos(q1 - pi/2)*cos(q3 - pi/2)*cos(q4 - pi/2)*sin(q2))*(cos(q1 - pi/2)*cos(q3 -
pi/2) - cos(q2)*sin(q1 - pi/2)*sin(q3 - pi/2)) + (150*cos(q2)*sin(q1 - pi/2)*sin(q4 -
pi/2) + 150*cos(q3 - pi/2)*cos(q4 - pi/2)*sin(q2)*sin(q1 - pi/2))*(cos(q3 - pi/2)*sin
(q1 - pi/2) + cos(q2)*cos(q1 - pi/2)*sin(q3 - pi/2)) + cos(q1 - pi/2)*sin(q2)*sin(q3
- pi/2)*(150*cos(q4 - pi/2)*(cos(q1 - pi/2)*sin(q3 - pi/2) + cos(q2)*cos(q3 - pi/2)
*sin(q1 - pi/2)) - 150*sin(q2)*sin(q1 - pi/2)*sin(q4 - pi/2)) + sin(q2)*sin(q1 -
pi/2)*sin(q3 - pi/2)*(150*cos(q4 - pi/2)*(sin(q1 - pi/2)*sin(q3 - pi/2) - cos(q2)*cos
(q1 - pi/2)*cos(q3 - pi/2)) + 150*cos(q1 - pi/2)*sin(q2)*sin(q4 - pi/2))) - q3dot*
((150*cos(q4 - pi/2)*(sin(q1 - pi/2)*sin(q3 - pi/2) - cos(q2)*cos(q1 - pi/2)*cos(q3 -
pi/2)) + 150*cos(q1 - pi/2)*sin(q2)*sin(q4 - pi/2))*(cos(q1 - pi/2)*sin(q3 - pi/2) +
cos(q2)*cos(q3 - pi/2)*sin(q1 - pi/2)) - (150*cos(q4 - pi/2)*(cos(q1 - pi/2)*sin(q3 -
pi/2) + cos(q2)*cos(q3 - pi/2)*sin(q1 - pi/2)) - 150*sin(q2)*sin(q1 - pi/2)*sin(q4 -
pi/2))*(sin(q1 - pi/2)*sin(q3 - pi/2) - cos(q2)*cos(q1 - pi/2)*cos(q3 - pi/2))) -
q4dot*((150*sin(q4 - pi/2)*(sin(q1 - pi/2)*sin(q3 - pi/2) - cos(q2)*cos(q1 - pi/2)
*cos(q3 - pi/2)) - 150*cos(q1 - pi/2)*cos(q4 - pi/2)*sin(q2))*(cos(q1 - pi/2)*cos(q3
- pi/2) - cos(q2)*sin(q1 - pi/2)*sin(q3 - pi/2)) - (150*sin(q4 - pi/2)*(cos(q1 -
pi/2)*sin(q3 - pi/2) + cos(q2)*cos(q3 - pi/2)*sin(q1 - pi/2)) + 150*cos(q4 - pi/2)
*sin(q2)*sin(q1 - pi/2))*(cos(q3 - pi/2)*sin(q1 - pi/2) + cos(q2)*cos(q1 - pi/2)*sin
(q3 - pi/2))))]
[
0,
-q1dot*cos(q1 - pi/2),
q2dot*cos(q2)*cos(q1 - pi/2) - q1dot*sin(q2)*sin(q1 - pi/2),

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q3dot*(sin(q1 - pi/2)*sin(q3 - pi/2) - cos(q2)*cos(q1 - pi/2)*cos(q3 - pi/2)) -
q1dot*(cos(q1 - pi/2)*cos(q3 - pi/2) - cos(q2)*sin(q1 - pi/2)*sin(q3 - pi/2)) +
q2dot*cos(q1 - pi/2)*sin(q2)*sin(q3 - pi/2)]
[
0,
-q1dot*sin(q1 - pi/2),
q1dot*cos(q1 - pi/2)*sin(q2) + q2dot*cos(q2)*sin(q1 - pi/2),
q2dot*sin(q2)*sin(q1 - pi/2)*sin(q3 - pi/2) - q3dot*(cos(q1 - pi/2)*sin(q3 - pi/2) +
cos(q2)*cos(q3 - pi/2)*sin(q1 - pi/2)) - q1dot*(cos(q3 - pi/2)*sin(q1 - pi/2) + cos
(q2)*cos(q1 - pi/2)*sin(q3 - pi/2))]
[
0,
0,
-q2dot*sin(q2),
q2dot*cos(q2)*sin(q3 - pi/2) + q3dot*cos(q3 - pi/2)*sin(q2)]

>>

```